Suggested Revisions for the North Dakota University Systems Funding Formula

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Executive Summary

While higher education has long been seen as one of the unchanging institutions of American society, trends in recent years bring this assumption into question. For instance, colleges and universities were quick to pivot instruction to online settings amidst the COVID-19 pandemic. Additionally, demographics and lowered student perceptions about the value of postsecondary education have led to decreases in student enrollment (Blake, 2024). As North Dakota's public institutions react with their own changes, it's essential that we evaluate their effects.

This report examines the current general funding formula used by the North Dakota University System (NDUS) and proposes revisions to better align the formula with the benefits to the state rather than costs only. The primary focus is on shifting the funding model from a cost-based approach to a comprehensive one that emphasizes the return on investment (ROI) for the state, while also considering the impact of in-person versus online education.

The existing NDUS funding formula, established for the 2013-15 biennium, is based on completed credit hours, incentivizing institutions to ensure students complete their courses. However, this formula does not fully account for the broader benefits of higher education to the state, such as increased economic output and tax revenues. We argue for a focus on the benefit generated from education, which would consider positive externalities, such as higher economic productivity.

The authors estimate the economic benefits of various degrees in North Dakota by analyzing wages, state income tax contributions, and worker retention rate (the percent of graduates who stay in the state). The analysis reveals higher online education rates correlate with lower worker retention varying by degree level. A Return-on Investment (ROI) analysis is also performed to show the return of NDUS funding to the state's economy. The results show bachelor's degrees yield the greatest returns per dollar followed by master's and doctoral degrees. Master's degrees offer comparable returns to bachelor's degrees when offered in-person, but lose over 20 percent of their value to the state when offered online due to lower worker retention. While associate degrees boast the greatest retention percentage of workers, their comparably lower gains to

wages and higher cost, in terms of NDUS funding, result in the lowest PV/C (present value to cost) ratios. Doctoral degrees, despite their high costs and low retention rates, generate high benefit to the state while contributing to scientific advancements. Although there are clear differences in outcomes across degree levels, all yield high investment returns to the state.

The report recommends revising the NDUS funding formula to consider benefits to the state rather than costs only, incorporating measures such as worker retention, wage increases, and tax contributions. A benefit-based funding formula would better signal to students and institutions which degrees are most beneficial to society, leading to more informed career choices and a stronger alignment between education and workforce demands in North Dakota.

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Introduction

To fund its public institutions, The North Dakota University System (NDUS) utilizes a funding formula which places a cost-based value on each completed credit (NDUS, 2024). In theory, programs are funded so long as students pass classes. Yet, the extent to which professors, students, and taxpayers understand this formula is limited. While costs may be useful for achieving stability in workforce education and program offerings, the current formula does not take the benefit to the state into account. This report details the structure of the most recent NDUS funding formula, analyzes the incentives it creates for institutions, estimates the benefits of online versus in-person degrees, and performs return on investment (ROI) calculations to estimate the social benefit of the most recently enacted NDUS funding formula by levels of degrees.

Current Funding Formula

The current NDUS funding formula is an altered version of what was adopted for the 2013-15 biennium. Senate Bill No. 2200 proposed a system that incorporated completed credits for the purpose of incentivizing institutions to help students successfully complete courses. While there have been changes to weighting, rates, and an elimination of an institution size factor, the formula structure remains largely the same (North Dakota University System, 2024).[1]

We use this current formula in our model to estimate cost. It does not include all costs incurred by students, the institution, or even the state of North Dakota (taxpayers), but simply the current credit-based funding formula. We are defining "cost" as how much public spending is contributed toward degree attainment through the funding formula. In short, we are analyzing the value of various degrees in terms of cost and benefit to the state of North Dakota.

Theory

Shift the Focus Toward Benefit Rather Than Cost

Justification for taxpayer-funded higher education lies in the social benefit education provides. Also known as positive externalities, increases in education lead to external benefits such as decreases in crime, increases in tax revenue, and increases in economic efficiency (Pigou, 1932). While an individual may earn a higher income or experience greater levels of fulfillment as the result of education, they don't fully account for these external social benefits. As a result, many economists agree that these types of goods, which have positive externalities, are underproduced without public or charitable funding.

[1] A description of this formula and tables representing how it is calculated can be found in the appendix.

"Welfare economics" informs how fiscal policy can be used to optimize societal wellbeing by increasing funding based on benefit, or, alternatively, taxing goods and services with social cost (Pigou, 1932). Yet, the current NDUS funding formula focuses exclusively on costs while allocating resources for a societal benefit. We fund certain degrees at various levels simply because they cost more. Further, the determination of these costs is opaque to the public, has remained largely unchanged since its inception, makes only an algebraic distinction between degree levels, and applies these antiquated cost estimates to all institutions regardless of institutional variation in cost. This likely leads to labor market distortions and exacerbates the issue of student education failing to match workforce demand. Students may create benefit for universities in terms of funding while failing to contribute towards North Dakota's GDP or tax revenue, both useful factors to estimate social benefit. Gross Domestic Product (GDP) measures the economic output of a region, reflecting the value of all goods and services produced. Education enhances GDP by boosting workforce productivity and earnings, leading to greater economic growth and tax revenues. If the funding formula considered benefit to the state, that would further signal to students which degrees are most beneficial to society and, likely, to lead to employment upon graduation. This is a crucial issue for North Dakota, as businesses have reported attracting and retaining qualified workers as the most important factor for future success (Bitzan, 2022).

Online Versus In-Person

Online course offerings increased following the COVID-19 pandemic. Yet, despite the prevalence of online degree options—even marketed to out-of-state students[2]—there is little data evaluating the extent to which these graduates impact state economies. There is evidence that employers view fully-online diplomas with greater skepticism (Fogle & Elliott, 2013), which could reflect in lower wages and lower employment rates. Some reasons for high drop-out rates among fully-online programs are students becoming too busy with work and family, not receiving support, difficult course material, and financial reasons (Sorenson & Donovan, 2017). On the other hand, Wayle and Ozogul (2019) found that student retention increases when students take at least one online course. While student retention towards degree completion is important, we have analyzed the effect of online degrees on North Dakota worker retention, defined as employment within the state following graduation.

It would seem some percentage of online degree offerings would benefit students and the state, while fully online offerings may reduce the state's student retention, employment, and wages. Another explanation could be that certain online degrees, such as business, allow professionals the ability to attain graduate degrees. These working professionals would therefore have a positive impact on worker retention and earnings given they are already working in a career and

[2] There is an offer of online programs at in-state tuition regardless of residency on UND's website (University of North Dakota, n.d.).

likely to stay in the state. For all students, course flexibility may be a factor in increased worker retention. It may provide avenues to work or pursue internships that wouldn't be possible in a less flexible environment.

Methodology

We use increases in wages and state income taxes while considering North Dakota worker retention (the percent of graduates who remain in the state) to provide a conservative estimate of economic benefit of various degrees in North Dakota.[3] The Census Bureau (n.d.) is adding a database to estimate postsecondary outcomes, but data is currently unavailable in North Dakota. We use Minnesota as a proxy to estimate the starting wage and percentage increase in wages for workers with various degrees in North Dakota across typical working ages.[4] We use the single filing schedule for state income tax (North Dakota Office of State Tax Commissioner, n.d.). North Dakota worker retention rates from North Dakota Information Technology and North Dakota Statewide Longitudinal Data System (n.d.) are used to estimate the percentage of workers remaining in North Dakota one year following graduation for the classes from 2010 to 2020. Given these data only provided rates, we use the Integrated Postsecondary Education Data System (IPEDS) to find the number of degrees attained by Classification of Instructional Programs (CIP) code and degree level from 2010 to 2020 (NCES, 2024a, 2024b). This also allows us to weight our data by number of degrees attained in the corresponding graduating class.

We calculate the costs associated with CIP codes and degree level by manually calculating the cost with a plan of study for the most popular degrees offered by NDUS for 2023 graduates[5] (Bismarck State College, 2024; Dakota College at Bottineau, 2024; Lake Region State College, 2024; North Dakota State College of Science, 2024; North Dakota State University, 2024; University of North Dakota, 2024; Williston State College, 2024; NDUS, 2023). These results are then aggregated using a weighted average.

[3] More details on how these aspects for the model were constructed can be found in the appendix.

[4] Minnesota PSEO data claims over 50% of graduates are sampled, which represents a more recent starting wage and 10 times the representative sample percentage of the PUMS dataset. The PSEO dataset uses median wages rather than mean, but median and mean wages are close at the beginning of a career and provide a conservative estimate (United States Census Bureau, 2024; US Census Bureau Center for Economic Studies, n.d.).

[5] For the two years in which UND provided a bachelor's for CIP 10, we found no degree plan. This graphic communication degree is similar to communication, so we used NDSU's communication degree as a proxy for cost and used communication wages.

The benefits and costs are then used to calculate present value (PV), net present value (NPV), the modified internal rate of return (MIRR), and the present value to cost ratio (PV/C).[6] PV measures the current value of future cash flows by discounting them at a rate that reflects the time value of money. NPV takes into account both the PV of cash inflows and outflows over time, providing a comprehensive assessment. MIRR is the discount rate at which the NPV of all cash flows equals zero, indicating the project's potential profitability; the project should only be undertaken if the MIRR meets or exceeds that of equally risky alternatives. PV/C shows how many times the PV of future cash flows exceeds the cost.

Using the percent of NDUS associate, bachelor's, master's, and doctoral graduates who work in North Dakota for NDUS graduates of 2010 to 2020, we run a regression to assess the impact of online degrees on the number of retained graduates in North Dakota for each degree level.[7] This regression is used to simulate the ROI of a fully online student versus a fully in-person student. Since the regression assumes the actual percent of retained workers and the actual percent of degrees that could be offered fully online for each degree level, we adjust the online percentage to be equal to 0 or 1, simulating the percent of workers who remain in the state for in-person (0) and online (1) programs. This new rate is then used to calculate a new ROI for a hypothetical online and hypothetical in-person student.

More detail regarding methodology and data use can be found in the appendix.

Results

Online Impact on Retained Workers Simulation

Table 1 displays the percent of college students who stay in North Dakota by degree level one year after graduating for graduates of 2010 to 2020. Additionally, Table 1 displays the probability that a fully online and fully in-person student would stay in North Dakota one year after graduation.[8]

[6] The modified internal rate of return is a conservative estimate of the return on investment in a project that more accurately considers typical returns on other state investments when projects yield large benefits in early years. We discount by the average U.S. cost of capital of 7.66% less expected inflation of 2.5% (Brigham & Houston, 2009; Creedy & Passi, 2018; Damodaran, 2024; Federal Reserve Bank of St. Louis, 2024).

[7] More details regarding the regression can be found in the appendix.

[8] Over 8.75 years, worker retention percentages do not change greatly. More detail regarding an alternative data set is described in the appendix. We used one year following graduation retention percentages for our simulation.

Table 1

PERCENT OF NDUS GRADUATES WHO STAY IN THE STATE AFTER GRADUATION

Degree Level	Percent Who Stay in the State†	Percent of Diplomas Available Online	Simulated Percent Who Stay in the State (100% Online)†	Simulated Percent Who Stay in the State (0% Online)†
Associate	56.84%	52.41%	52.17%	61.99%
Bachelor's	41.90%	25.03%	38.82%	42.93%
Master's	41.25%	50.07%	35.83%	46.66%
Doctoral	37.20%	4.87%	19.71%	38.09%

† Weighted by the number of graduates in each CIP.

As the table shows, as higher-level degrees are attained, worker retention decreases, conveying higher worker mobility. It should be noted that this does not mean North Dakota has little demand for higher-level degrees. We do not have data regarding workers entering the state from elsewhere who may also have higher-level degrees. So, it may be that there is greater mobility among all states for these workers. Yet, this does indicate that the graduates who receive the most per-credit support from the state tend to leave at the highest rates.

The table also shows that higher online rates lead to fewer students staying in the state. Since the simulation of online or in-person enrollment only affects the worker retention component of the ROI model, costs are equal, representing the current spending by North Dakota according to the most recent funding formula. Yet, benefits are impacted by this change in worker retention; when fewer workers remain in the state, fewer benefits to state GDP and taxes occur.

We do not analyze variation in wages as a result of an online degree. We only estimated the impact of online degree rates on retention. So, if there is a negative impact on wages, the ROI would be comparatively lower for an online student versus an in-person student.

ROI Model

ROI Inputs Analysis

In order to estimate ROI by degree level, we first calculate the benefits for all CIP and degree combinations, aggregating them at the degree level using degrees awarded data for NDUS institutions from the National Center for Education Statistics (NCES, 2024a, 2024b). The benefits include increase in wages and tax revenues resulting from attaining various degrees. The increase in wages is used as a proxy for the increase in productivity and consequently in GDP. The simple logic is that employers are only willing to pay higher wages to employees if they

generate at least that much value to the business. Associate degree benefits and bachelor's degree benefits are calculated as the average wage and income tax generated above that of a high school diploma, weighted by the number of graduates within each CIP. Master's degree benefits are calculated based on the increases above a bachelor's degree. Doctoral or professional degree benefits are calculated as the increases above a master's degree, except for legal and medical, which are increases above a bachelor's degree.[9] Since degrees are attained at different ages, and more education requires the sacrifice of wages that could have been earned while working, we began the career of an associate graduate, bachelor's graduate, master's graduate, and doctoral graduate at 20, 22, 24, and 26 years of age respectively. We assume their careers continue until the end of age 66. Table 2 displays the increased wages by degree level at various ages over the course of a career, weighted by the number of graduates in each CIP. Table 3 displays the additional revenue generated in state income taxes.

Table 2

Wage Increases From Attaining Various Degrees at Selected Ages (In 2023 USD)

Degree Level	First Year After Graduation†	Age				
		30	40	50	60	66
Associate††	$11,030	$9,631	$9,837	$11,650	$15,067	$17,889
Bachelor††	$14,463	$42,173	$65,262	$75,518	$72,942	$65,236
Master†††	$10,913	$33,472	$59,020	$69,505	$64,928	$54,952
Doctor††††	$27,759	$58,119	$114,036	$141,407	$140,233	$125,826

† Age 20, 22, 24, 26 for associate, bachelor's master's, and doctoral, respectively.
†† Relative to high school degree.
††† Relative to bachelor's degree.
†††† Relative to master's degree except for MD and JD.

Table 3

Additional Taxes Collected From Those With Various Degrees at Selected Ages (In 2023 USD)

Degree Level	First Year After Graduation†	Age				
		30	40	50	60	66
Associate††	$59	$188	$192	$227	$294	$349
Bachelor††	$153	$822	$1,273	$1,473	$1,422	$1,272
Master†††	$212	$653	$1,193	$1,444	$1,337	$1,101
Doctor††††	$539	$1,130	$2,247	$2,953	$2,905	$2,515

[9] Additionally, a medical degree is in relation to a biology bachelor's, since that degree is more common than a bachelor's of nursing for North Dakota and Minnesota medical doctors. The law doctoral degree wages are relative bachelor's in pre-legal studies.

† Age 20, 22, 24, 26 for associate, bachelor's, master's, and doctoral, respectively.
†† Relative to high school degree.
††† Relative to bachelor's degree.
†††† Relative to master's degree except for MD and JD, which are relative to bachelor's.

Associate degrees lead to early and sustained wage increases above that of a high school degree alone. Yet, other degrees quickly outpace the earnings of an associate degree, with doctoral degrees resulting in the greatest gains. Master's degrees, which may be half the credits of associate degrees and a quarter of the credits of bachelor's degrees yield large gains to wages. It may be that, since many working professionals attain a master's degree after gaining some years of experience, that their higher wages are a result of a combination of education and experience rather than degree attainment alone. Further research would need to be completed to determine the degree to which experience or a master's degree are responsible for these increased wages.

Considering the Percent of Degree Recipients Who Remain in the State

The tables above show an estimate of the increase in earnings (GDP proxy) and tax collections from an individual worker who earns various degrees in North Dakota. However, in order to estimate a return on investment to funding such a degree through the funding formula, we need to consider the likelihood that the worker will stay in the state after graduation. If the student leaves, the ensuing productivity gains from the degree are never realized by the state. We estimate the likelihood that the worker will stay in the state by the percent of students achieving that degree who stay in the state one year following graduation. Table 4 multiplies the sum of increased earnings and tax receipts of achieving various degrees by the percent who stay in the state. It should be noted that this is a very conservative of the benefit of these degrees to the state since many of these workers would not be in the state if they were not pursuing a degree here; that is, the benefit of attracting new workers is not accounted for.

Table 4

Benefit of Individual Degrees to the State at Select Ages – Considering Percent Who Stay in the State Upon Graduation (In 2023 USD)

Degree Level	First Year After Graduation†	Ages				
		30	40	50	60	66
Associate††	$6,531	$5,917	$6,107	$7,194	$9,178	$10,800
Bachelor††	$6,184	$18,130	$28,037	$32,437	$31,330	$28,022
Master†††	$5,236	$14,778	$25,601	$30,058	$28,130	$23,907
Doctor††††	$9,244	$20,127	$40,220	$50,178	$49,886	$44,791

† Age 20, 22, 24, 26 for associate, bachelor's, master's, and doctoral, respectively.
†† Relative to high school degree.
††† Relative to bachelor's degree.
†††† Relative to master's degree except for MD and JD, which are relative to bachelor's.

As our table shows, due to the higher state retention rates for associate degrees, bachelor's degrees often yield lower benefit than associate degrees early on despite associate degrees attainment occurring at a younger age. Master's and doctoral degree holders can earn significantly more than those with associate or bachelor's degrees, particularly over a longer period, highlighting the long-term financial returns of advanced education.

Table 5 displays the degree cost[10], MIRR, present value (PV), PV-to-cost ratio (PV/C), and net present value (NPV) using weighted averages of CIP code benefits.

Table 5

RETURN ON INVESTMENT FOR VARIOUS DEGREES FROM FUNDING FORMULA MONEY

(IN 2023 USD)

Degree Level	Cost	MIRR	PV	PV/C	NPV
Associate	$15,909	9.67%	$114,364	7.19	$98,454
Bachelor's	$17,766	12.49%	$368,432	20.74	$350,667
Master's	$18,370	12.54%	$340,036	18.51	$321,667
Doctoral	$51,434	11.50%	$566,636	11.02	$515,203

The ROI factors corroborate our findings that bachelor's and master's degrees provide greater benefits to North Dakota's local economy, yielding high wages and average retention rates for comparably low cost.

The ROI factors reveal that all levels of education provide significant benefit to the state. While associate degrees offer the lowest PV/C and MIRR, these investment metrics are still impressive (with benefits exceeding costs by seven times and an inflation-adjusted annual return of nearly 10 percent). Bachelor's degrees offer a MIRR just below master's and the highest PV/C, indicating the best return for each dollar invested in education (with benefits exceeding costs by more than 20 times). Master's degrees boast the highest MIRR and the second-highest PV/C. Doctoral and professional degrees, despite their high benefit increases, have the highest average cost, resulting in lower MIRR and PV/C ratios compared to bachelor's and master's degrees. Yet, they should not be overlooked due to their scientific contributions and substantial absolute returns. While there are clear differences in educational investments, all yield impressive returns. For instance, the S&P 500, a popular investment index, has annual real return rate of 8.18 percent over past 40 complete years and has substantially more risk (Damodaran, 2024).

It's especially true that education has been a good investment for North Dakota considering our estimates are conservative. In order for an employer to be willing to pay a given wage to an

[10] NDSU and UND rates were used for bachelor's, master's, and doctoral cost calculations. Yet, most associate degrees come from smaller institutions. We used NDSCS funding figures for our cost calculations of associate degrees.

employee, that employee must generate a profit greater than their wage. Since we do not have data on the GDP generated by every student, we are using wages as a conservative proxy. Additionally, we are not accounting for students who are from out of state and attain associate, bachelor's, or master's degrees in North Dakota and stay. In this scenario, North Dakota receives the benefit of all their schooling, not just of the most recent publicly subsidized degree. By funding these degrees and students, the actual impact on ROI would be much greater.

Sensitivity Analysis

We conduct a fixed-effect ordinary least squares (OLS) regression to simulate how changes in online course rates could influence the percent of graduates staying the state. Tables 6 and 7 use these estimates, found in Table 1 to show the benefits to the state of individual online and in-person degrees.

Table 6

Simulated Benefit of Individual Degrees to the State at Select Ages for In-Person Degrees (In 2023 USD)

Degree Level	First Year After Graduation	Ages				
		30	40	50	60	66
Associate	$7,122	$6,452	$6,659	$7,845	$10,009	$11,777
Bachelor's	$6,336	$18,575	$28,725	$33,233	$32,099	$28,710
Master's	$5,923	$16,717	$28,959	$34,001	$31,819	$27,043
Doctoral	$9,466	$20,611	$41,189	$51,386	$51,087	$45,869

Table 7

Simulated Benefit of Individual Degrees to the State at Select Ages for Online Degrees (In 2023 USD)

Degree Level	First Year After Graduation	Ages				
		30	40	50	60	66
Associate	$5,995	$5,431	$5,605	$6,603	$8,425	$9,913
Bachelor's	$5,729	$16,798	$25,977	$30,053	$29,027	$25,963
Master's	$4,551	$12,845	$22,252	$26,126	$24,450	$20,780
Doctoral	$4,897	$10,663	$21,308	$26,584	$26,429	$23,730

Bachelor's degrees are most resistant to decreases in benefits as a result of online delivery of courses. Consistent with our findings that larger programs may actually increase retention, these bachelor's degrees likely experience this effect for some degrees. This is likely due to the fact that it is rare for a student to fully complete a bachelor's degree online. The availability of some online courses may benefit student retention and worker retention. Alternatively, associate and master's degrees may be fully completed online and are increasingly tailored to this method of

delivery. This is likely leading to out of state students attaining these degrees while staying out of state, decreasing state retention rates for these programs. Yet, if this flexibility allows working professionals to pursue degrees, who have a high chance of staying within the state, this is likely to benefit state retention. Student-level data on residence would be needed to explore these opposing effects in greater detail. Doctoral and professional degree worker retention is most sensitive to online rates, but the current online rate is slightly below five percent, so these large simulated decreases are less reliable for doctoral degrees.

Table 8 displays the ROI metrics for a simulated in-person and online student at various degree levels.

Table 8

Simulated Return on Investment for Various Degrees From Funding Formula Money

(In 2023 USD, Online = 100% Online Rate, In-Person = 0% Online Rate)

Degree Level	Percent Staying in ND After Graduation		MIRR		PV Benefits		PV/C		NPV	
	Online	In-person	Online	In-person	Online	In-person	Online	In-person	Online	In-person
Associate	52.17%	61.99%	9.47%	9.87%	$104,970	$124,710	6.60	7.84	$89,061	$108,800
Bachelor's	38.82%	42.93%	12.30%	12.55%	$341,353	$377,474	19.21	21.25	$323,587	$359,708
Master's	35.86%	46.66%	12.18%	12.87%	$295,553	$384,640	16.09	20.94	$277,184	$366,270
Doctoral	19.71%†	38.09%	9.78%†	11.56%	$300,200†	$580,282	5.84†	11.28	$248,766†	$528,848

† Since the actual percentage of online degrees available for doctoral degrees is only 4.87 percent, our simulation is less reliable in estimating the effects of an online rate of 100 percent. Additionally, doctoral observations are comparably low.

The analysis reveals that bachelor's degrees are least affected by the mode of delivery, with benefit-cost ratio dropping from 21 for in-person delivery to over 19 for online delivery. Associate degrees reveal a benefit-cost ratio of nearly 8 for in-person delivery and 6.5 for online. Master's degrees reveal the greatest meaningful sensitivity to method of delivery with benefit-cost ratio dropping from 21 to 16 when comparing in-person and online delivery. While the difference is greatest for doctoral and professional degree holders, online delivery only represents 4.87 percent of degrees, so the effect of online delivery on retention demands less confidence.

Funding Formula With Benefits vs. Costs

At the beginning of this study, we noted the importance of considering benefits to the state in the funding formula. The current funding formula awards funds to higher education institutions based on the relative costs of providing completed credits. Table 9 shows the relative ratio of funds provided for a credit of various degree levels in comparison to an online bachelor's using the current cost-based formula and a benefits-based formula. The ratios in the cost-based formula are calculated by obtaining weighted average funding formula amounts per credit across all CIPs (where weights are the number of degrees in each CIP). The ratios in the benefit-based formula are calculated using present values of benefits by degree. The difference in these columns, or ratios by degree, reveals the difference in a benefit-based formula versus the one currently employed by NDUS. Additionally, this table highlights how the current funding formula fails to differentiate funding based on mode of delivery despite differences in benefits.

Table 9
COMPARISON OF FUNDING FORMULA CREDIT WEIGHTS
BASED ON BENEFIT RATIOS VERSUS COST RATIOS

Degree and Delivery	Benefit Ratios (to Online Bachelor's)	Current Formula Cost Ratios (to Online Bachelor's)
Online Associate	0.308	0.895
Online Bachelor's	1.000	1.000
Online Master's	0.866	1.034
Online Doctoral	0.879[†]	2.895
In-Person Associate	0.365	0.895
In-Person Bachelor's	1.106	1.000
In-Person Master's	1.127	1.034
In-Person Doctoral	1.700	2.895

[†] Since the actual percentage of online degrees available for doctoral degrees is only 4.87 percent, our simulation is less reliable in estimating the effects of an online rate of 100 percent. Additionally, doctoral observations are comparably low.

In comparing these columns, in-person bachelor's and master's programs would receive more funding if based on benefits, while online programs would receive less. Although associate degrees provide clear benefits to the state, their funding based on costs appears to be much higher than if they were funded based on benefits.

Conclusion

This report concludes that the current funding formula for the North Dakota University System (NDUS) requires significant revisions to better align with the state's economic goals and workforce needs. The following key points summarize the findings and recommendations: 1. Higher rates of online degrees correlate with lower state worker retention rates. Associate and master's degrees are more sensitive to the delivery methods. A fully online degree has approximately 10 percentage points lower state retention than a fully in-person degree for both of these degree levels. 2. When ranking by economic benefits from increased wages and tax revenues, bachelor's and master's degree have the highest metrics. Associate degrees currently have a comparably high base funding relative to wage increases. Doctoral degrees, despite their high costs, provide substantial long-term economic gains while contributing to scientific advancements. 3. In-person education enhances the retention rate and benefits to the state, especially for associate and master's degrees.

To optimize the economic benefits of higher education for North Dakota, we proposed following revisions to the NDUS funding formula: Shift from a cost-based to a hybrid funding model that incorporates both costs and benefits; differentiate funding for online and in-person education for

bachelor’s degrees, and especially for associate, master’s and doctoral degrees; implement a transparent and flexible funding formula that adapts to changes in education and workforce needs, providing clearer signals to students and institutions regarding the most valuable degrees for the state's economy.

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Appendix

Current Funding Formula

Tables A1, A2, and A3 display the inputs for funding given to each institution's base funding. While there are funds allocated for special programs or projects, this is the main formula by which North Dakota contributes to the funding of NDUS institutions. Tables A1 and A2 show the base rate and credit completion factor, both of which benefit smaller institutions. Table A3 represents the credit weight, which is a multiplier for the level of credits and degree of credits based on cost.

Table A1

BASE RATES FOR NDSU INSTITUTIONS

Institution	Base Rate 2021-23
Bismarck State College	$110.38
Dakota College at Bottineau	$110.38
Lake Region State College	$110.38
North Dakota State College of Science	$110.38
Williston State College	$110.38
Dickinson State University	$103.76
Mayville State University	$103.76
Minot State University	$103.76
Valley City State University	$103.76
North Dakota State University	$73.15
University of North Dakota	$73.15

Table A2

CREDIT COMPLETION FACTOR

Factor Value	Minimum Credits for Threshold
1	240000
1.05	180000
1.1	135000
1.15	130000
1.2	120000
1.3	70000
1.4	60000
1.5	50000
1.6	40000
1.7	30000
1.8	0

Note: "if the number of credit-hours completed by an institution would cause a decrease in the credit completion factor for that institution compared to the credit completion factor the institution was entitled to receive during the 2017-19 biennium, the state board of higher education shall multiply the product determined under section 15-18.2-02 by a credit growth factor of 1.0 for credits in excess of the factor the institution was entitled to receive during the 2017-19 biennium. Any weighted credit-hours multiplied by a credit growth factor under this section must have a weight of 1.0 under section 15-18.2-02. The state board of higher education shall multiply the remaining weighted credits by the credit completion factor the institution was entitled to receive during the 2017-19 biennium" (North Dakota Legislature, n.d.).

Table A3

CREDIT COMPLETION WEIGHTS

Course Level	Agriculture, Transportation, Biological and Physical Sciences, Business, and Education	Architecture	Engineering	Health Sciences	Legal Studies	Core	Career and Technical Education	Basic Skills
Lower (100-200)	1.9	1.8	2.5	3	3.5	1	5	2.3
Upper (300-400)	3.8	3.6	5	6	7	2	5	2.3
Professional	5.7	5.4	7.5	9	14	3	5	2.3
Graduate	7.6	7.2	10	12	14	4	5	2.3
Medical School				34.5			5	2.3

Each biennium, the completed credits for each institution determine funding for the next biennium. Each completed credit is multiplied by the corresponding value in Table A3. The total number of these completed credits are used to determine the credit completion factor, or CCF, shown in Table A2. That CCF is multiplied by the value of all credits determined by Table A3 and the corresponding base rate of Table A1. This formula is represented by Equation 1.

$$Funding = BaseRate_i \times CreditCompletionFactor_i \times \left(\sum_l \sum_c CreditWeight_{lc} \times CompletedCredits_{ilc}\right) \quad \text{(A1)}$$

In the formula, *i* denotes institution, *l* denotes levels of credit hours, and *c* denotes CIP code.

Credit Completion Factor Suggested Reforms

The shortcoming of the current credit completion factor (CCF) is its use of thresholds.[11] Originally, the steps were simpler. Each CCF step occurred at a fixed increase in completed credits. Yet, it still created potential scenarios in which institutions could lose funding by increasing completed credits (Dolan, 2014). While it's difficult to say whether this system has led to adverse decision-making, which would maximize institutional funding at the expense of student success, the current formula incentivizes that scenario. As an example, according to the January 26th, 2024 Funding Formula Study Group PowerPoint, Dickinson State University (DSU) had 59,541 credit hours completed in 2015-17, 60,655 in 2017-19, and 60,000 in 2019-21 (NDUS, 2024). Since the 60,000th credit completed moved the funding threshold of DSU to receive a CCF multiplier of 1.4 rather than 1.5, that passed credit cost DSU $1,313,153 in funding.[12] In recent biennia, this problem has been mitigated by allowing institutions which would decrease funding to keep their previous CCF with all additional credits multiplied by a credit growth factor of 1 (NDUS, 2024). If such a provision was not made, should the administrators of DSU put efforts towards maximizing funding, they could have reduced course offerings, enrollments, or encouraged policies to lower passing rates to achieve more funding.

[11] As an example, a threshold issue exists today following the Affordable Care Act requirement that health insurance be provided to employees of companies with 50 or more full-time employees. The ratio of part time employees increased and many companies capped employment at 48 or 49 employees. The reason is that the 50th employee didn't only add the cost of medical insurance for themselves, but also for the existing 49 employees. If businesses choose not to provide insurance but chose to hire the 50th employee, they have to pay a tax penalty (Tanner, 2012; Cigna Healthcare, 2024). While businesses may be more efficient at responding to these incentives due to their goal of maximizing profit, it seems likely such a scenario may affect institutional decisions.

[12] This was the marginal cost of that last credit. It was calculated by taking what DSU was awarded in base funding, $18,393,063 divided by their 1.4 CCF and multiplied by the CCF, 1.5 that they would have had with 59,999 credits completed. We then subtracted the reward that student would have generated, approximately $637, assuming a weighted credit hour of 5. The difference between revenue at 59,999 credits and 60,000 credits is therefore $1,313,153.

Since completed credits have been near the threshold for DSU the past three biennia, creating a more permanent solution would benefit planning for the state and institutions.

This issue of institutions being potentially penalized for increasing enrollment was brought up in 2019 (*Higher Education Funding Formula Review Committee*, 2019). This goes against the intended purpose of the CCF to encourage performance of institutions by ensuring students are successfully attaining credits. We propose three solutions, which offer various algebraic alternatives to the threshold model. With one exception, they all prevent the scenario in which some completed credit hours immediately after each threshold reduce funding. The current threshold system is visualized in Figure A1.[13] Figure A2 almost solves this problem by establishing new rates, rather than thresholds, between each threshold value, created by calculating a slope between each value. In this model, given current threshold rates, there is one section which remains problematic. Alternatively, a simple linear model with a minimum CCF is displayed in Figure A3. It should be noted that this model, by not accounting for the dip in CCF for middle values, is more generous than all other figures. Figure A4 uses an exponential model which closely resembles Figure A1. Figure A4, used a polynomial to create a formula similar to the current rates. It is our recommended model. While Figures A2, A3, and A4 affect institutions differently, they all reduce or prevent the threshold problem caused by the current CCF threshold formula.

[13] Figure 1 does not account for the recent provision that prevents a reduction in funding.

Figure A1

CURRENT THRESHOLD FUNCTION

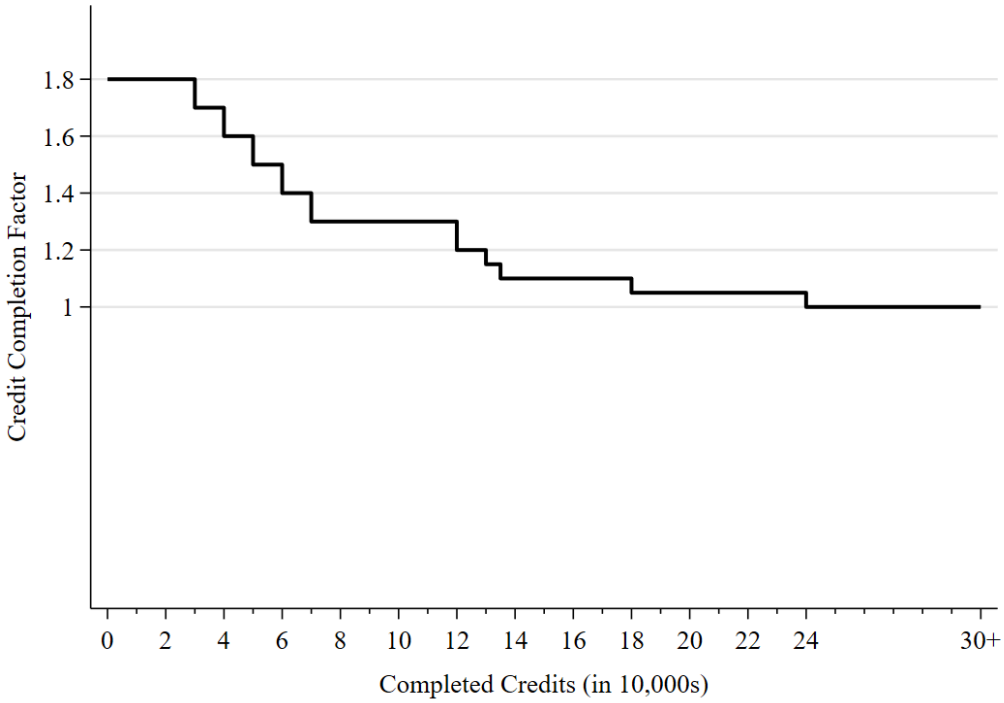


Figure A2

CURRENT SYSTEM WITH LINES BETWEEN VALUES

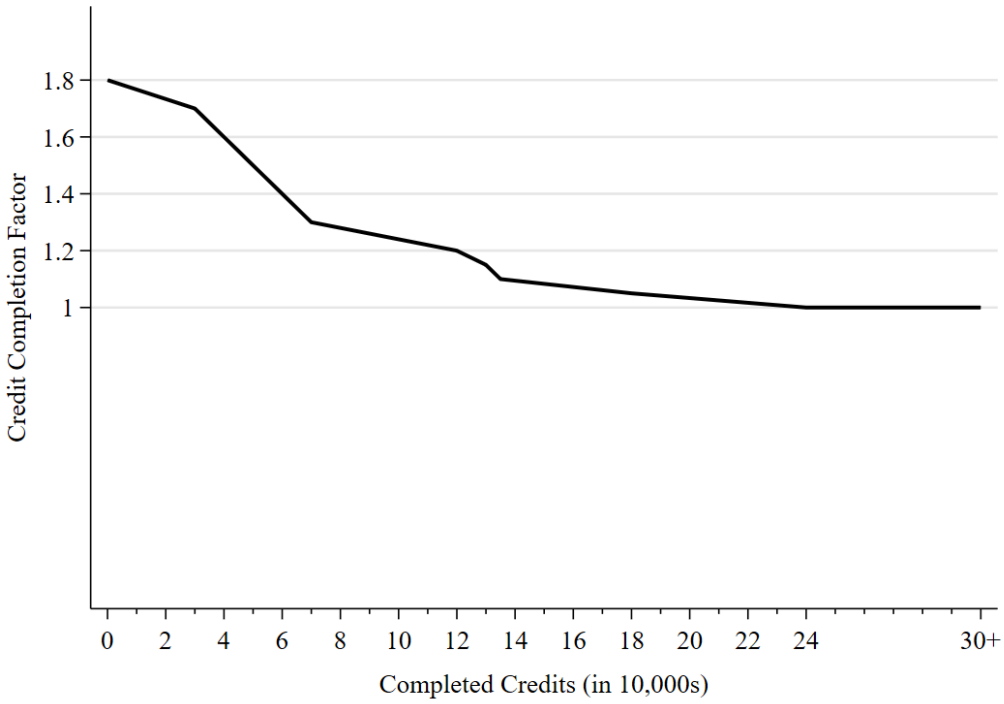

Credit Completion Factor
1.8
1.6
1.4
1.2
1
0
2
4
6
8
10
12
14
16
18
20
22
24
30+
Completed Credits (in 10,000s)

Figure A3

SIMPLE LINEAR FUNCTION

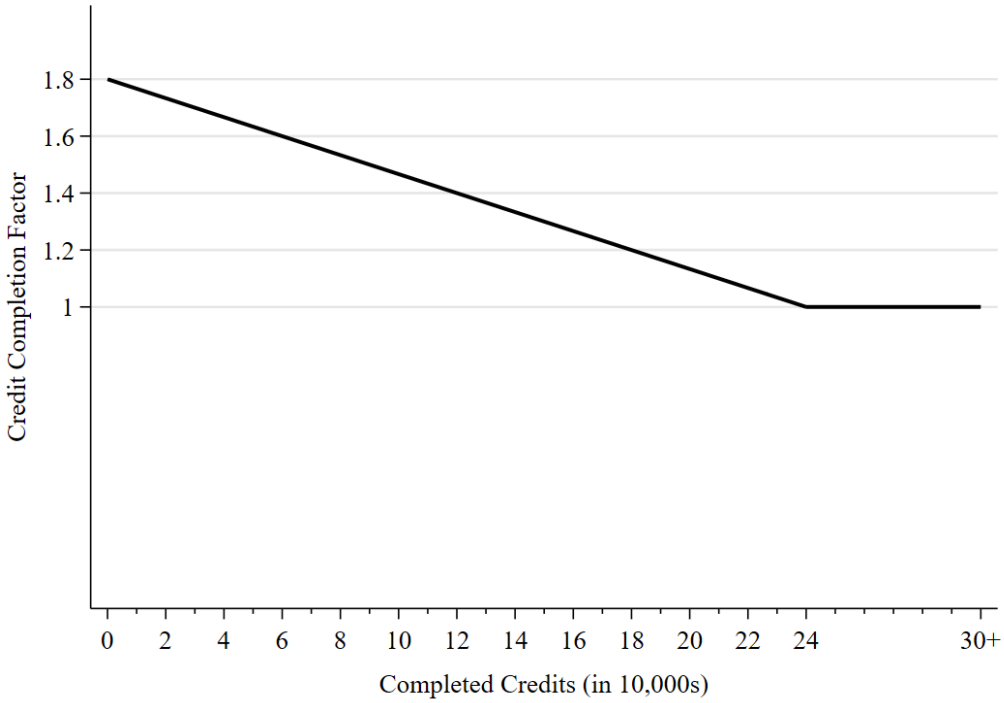
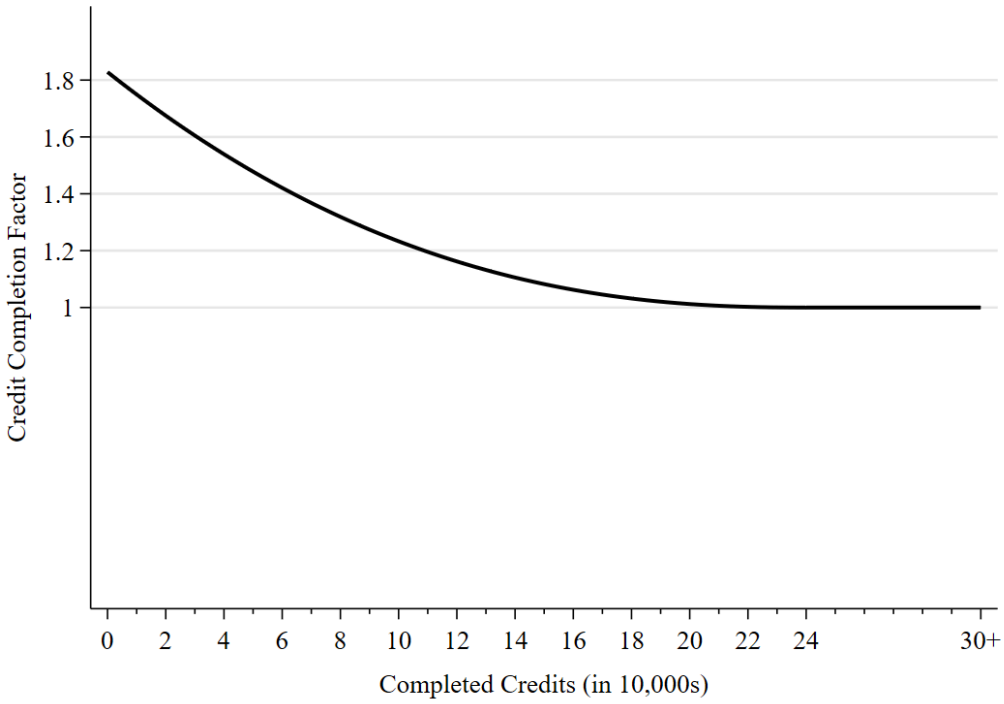


Figure A4

EXPONENTIAL FUNCTION


Credit Completion Factor
1.8
1.6
1.4
1.2
1
0
2
4
6
8
10
12
14
16
18
20
22
24
30+
Completed Credits (in 10,000s)

To determine the total effect that each formula would have on funding, Figures A5-A8 show the result of multiplying the CCF by Completed Credits for each function represented in Figures A1-A4. This shows the total multiplier that an institution would receive at various levels of credit completion. Any negative slope for these graphs shows how an increase in completed credits would reduce funding. This occurs at every change in Figure A5 and at one point in Figure A6. There is no occurrence for Figures A7 or A8, which represent a linear function and an exponential function respectively.

The specification that institutions which would decrease their CCF by increasing students would maintain their previous CCF and all additional credits be multiplied by a credit growth factor of 1.0 is visualized by the dotted lines in Figure A5. Yet, this still leads to strange distortions and confusion, especially when considering schools which may continue to increase in the next biennium and still lose funding.

Figure A5

CURRENT THRESHOLD FUNCTION TOTAL EFFECT

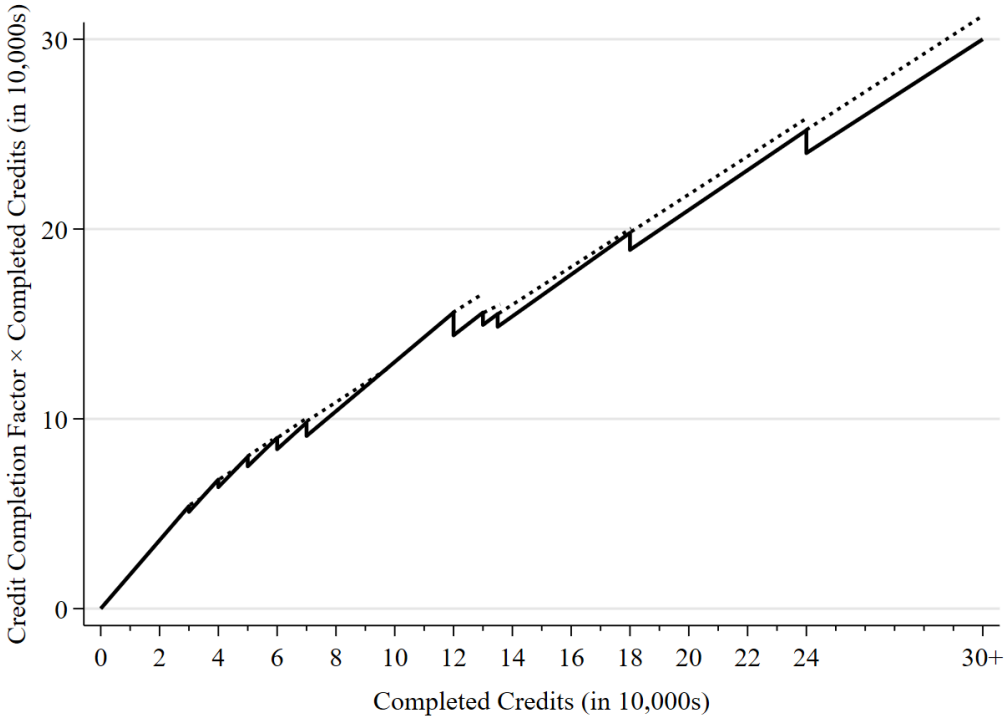


Figure A6

CURRENT SYSTEM WITH LINES BETWEEN VALUES TOTAL EFFECT

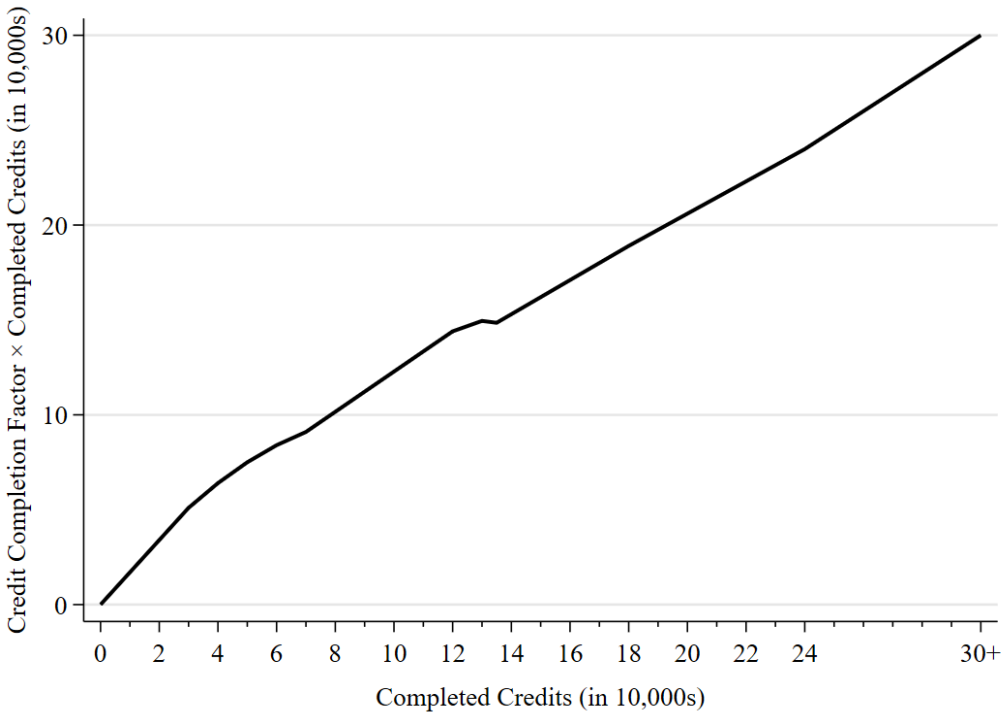


Figure A7

SIMPLE LINEAR FUNCTION TOTAL EFFECT

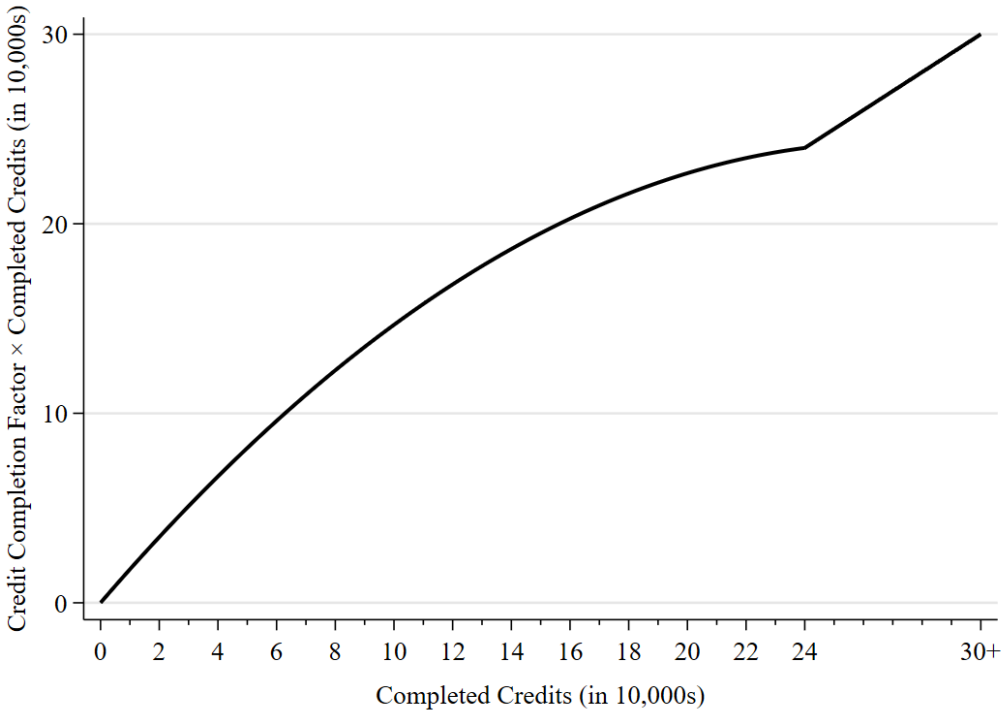


Figure A8

EXPONENTIAL FUNCTION TOTAL EFFECT

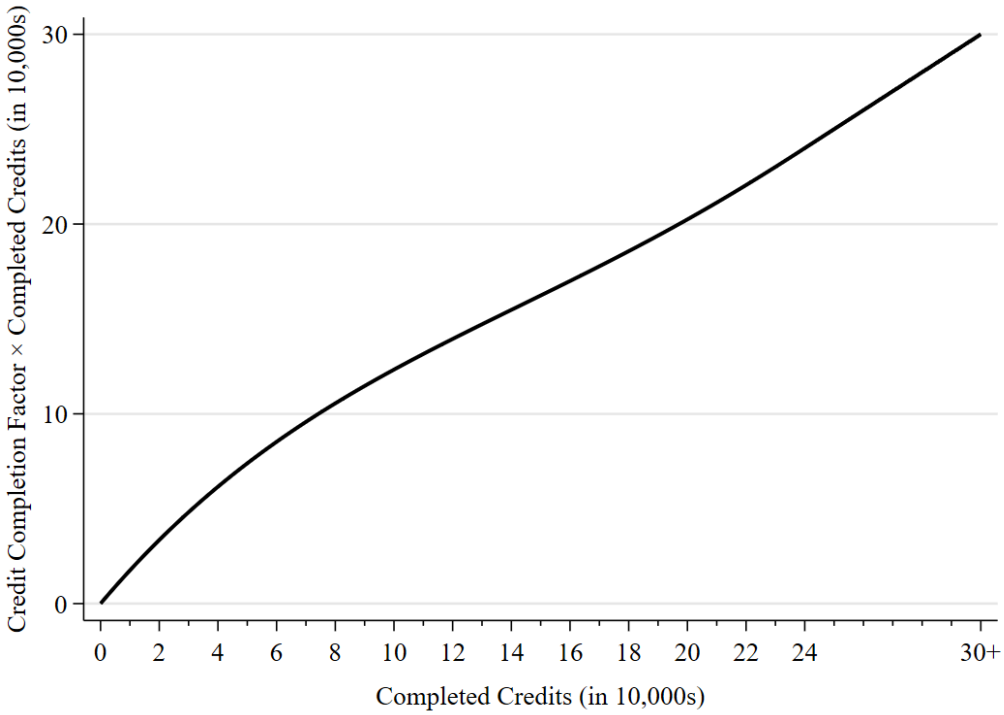


Figure A6 reveals that funding decreases for one section of the graph. Figure A7 clearly shows increased funding for mid-sized institutions. Figure A8, our formula of choice, displays a smooth line that closely resembles Figure A5 without causing decreased funding as completed credits increase.

Table A4

FORMULAS FOR FIGURES A1-A8

Figure A…	Formula
1)	Calculated using the current threshold relationship for each pair of values of CCF and Completed Credits. Dotted line represents calculation in the scenario that funding would be lower from last biennium.
2)	Calculated by creating a line between each pair of values of CCF and Completed Credits.
3)	If $CompletedCredits \geq 240000,\ CCF = 1$ If $CompletedCredits < 240000,\ CCF = -\frac{CompletedCredits}{300000} + 1.8$
4)	If $CompletedCredits \geq 240000,\ CCF = 1$ If $CompletedCredits < 240000,\ CCF = 1 + \left(\frac{240000 - CompletedCredits}{260000}\right)^{\frac{40}{17}}$
5)	Same as Figure 1, but multiplies each CCF by Completed Credits.
6)	Same as Figure 2, but multiplies each CCF by Completed Credits.
7)	If $CompletedCredits \geq 240000,\ CCF = CompletedCredits$ If $CompletedCredits < 240000,\ CCF = \left(-\frac{CompletedCredits}{300000} + 1.8\right)CompletedCredits$
8)	If $CompletedCredits \geq 240000,\ CCF = CompletedCredits$ If $CompletedCredits < 240000,$ $CCF = \left(1 + \left(\frac{240000 - CompletedCredits}{260000}\right)^{\frac{40}{17}}\right)CompletedCredits$

Online Student Simulation and Retention

We used a fixed-effects OLS regression to estimate the correlation of online education and worker retention.

$$RetainedWorkers_{cy} = \beta_0 + \beta_1 OnlineRate_{cty} + \beta_2\left(OnlineRate_{cty} \times TotalDegrees_{cty}\right) + \beta_3 TotalDegrees \quad \text{(A2)}$$

$RetainedWorkers_{cy}$ represents the estimated total number of retained workers. This is done by multiplying the total number of degrees attained by CIP code, degree type, and year by the corresponding state retention rate from SLDS (Statewide Longitudinal Data System) data. $OnlineRate_{cty}$ represents the percentage of degrees available online from NDUS institutions by CIP code, degree type, and year. $TotalDegrees_{cty}$ represents total number of degrees attained by CIP code, degree type, and year. The interaction terms are included to capture the different relationships between online percentage and workers retained for different sized programs and degree levels. This model includes fixed effects by CIP code, ν_c, and year, ω_y, meaning these differences do not drive the correlation results. Lastly, μ_{cty} represents an error term.

If graduates do not continue to work in North Dakota, they much less significantly benefit the state in terms of GDP and tax revenues. Therefore, we use North Dakota worker retention rates from North Dakota Statewide Longitudinal Data System (2019) to estimate the percentage of workers remaining in North Dakota following graduation for the classes of 2010 to 2020. This data both measures worker ability to become employed, but also stay employed in North Dakota. Given this data only provided rates, we use the Integrated Postsecondary Education Data System (IPEDS) to find the number of degrees attained by CIP code and degree level from 2010 to 2020. This also allows us to weight our data by number of degrees attained in the corresponding graduating class. Lastly, IPEDS data includes whether a degree can be completed fully online. Institutions simply select whether this is possible for a given degree. It does not indicate the number of online courses actually conducted or how many online courses a student has taken. This does mean that a student could be counted as part of our online rate while never taking an online class. While more specific data would be ideal, such as student level online enrollment and outcome data, this serves as a useful proxy for online enrollment.

SLDS also provides an 8.75-year estimate of worker retention in the state, but the pools of graduates are unknown, and the reliability of the data is questionable. Therefore, we selected the workers retained one year after graduation for our estimate.

Table A5 displays the results of the OLS regression.

Table A5

EFFECT OF ONLINE RATES ON RETAINED WORKERS

	OLS
Online %	-24.09***
	(7.598)
Online % × Degrees	0.0676*
	(0.0347)
Degrees	0.458***
	(0.0175)
Online % × Bachelor's	-4.732
	(9.860)
Online % × Master's	2.048
	(8.224)
Online % × Doctoral	-3.787
	(12.25)
Bachelor's	-24.78***
	(4.767)
Master's	-22.95***
	(5.499)
Doctoral	-45.45***
	(8.142)
Constant	17.98***
	(5.285)
Observations	455
R-squared	0.976

CIP and year fixed effects were not included in the table.
Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Equations A3-A6 use parameters from Table A5 to estimate changes in state worker retention with changes in online rates for various degrees. The interaction term between online percent and degrees shows that larger programs can maintain online offerings while still providing connection for students. The other interaction terms show how online percentages affect in state retention differently for different types of degrees.

$$\frac{\partial RetainedWorkers_{associates}}{\partial Online\%} = -\ 24.08768\ +\ 0.0676088 Degrees \quad \text{(A3)}$$

$$\frac{\partial RetainedWorkers_{bachelors}}{\partial Online\%} = -\ 24.08768\ -\ 4.732265\ +\ 0.0676088 Degrees \quad \text{(A4)}$$

$$\frac{\partial RetainedWorkers_{masters}}{\partial Online\%} = -\ 24.08768\ +\ 2.048139\ +\ 0.0676088Degrees \quad \text{(A5)}$$

$$\frac{\partial RetainedWorkers_{doctorate\&professional}}{\partial Online\%} = -\ 24.08768\ -\ 3.787389\ +\ 0.0676088Degrees \quad \text{(A6)}$$

Table A6 displays the actual online rate, average number of degrees by CIP, estimated retained workers, and the state retention percentage of workers. We use the average size of degrees for each type of degree to calculate the actual number of retained workers by degree level at the actual online rate. We then set the online rate to 0 and 1 and simulate the change in retained workers as a result of these changes in the online rate, as shown in Table 8.

Table A6

SIMULATED RETENTION RATE WITH ACTUAL ONLINE RATES

	Online Rate	**Average Degrees per CIP**	**Retained Workers**	**Retention Percentage**
Associate's	52.41%	145.35	82.62	56.84%
Bachelor's	25.03%	221.62	92.86	41.90%
Master's	50.07%	137.11	56.56	41.25%
Doctoral	4.87%	95.79	35.63	37.20%

For large degrees, with many graduates, our simulation indicates a possible increase in retention. Yet, almost all degrees are smaller than is necessary for this positive effect. It's worth noting that the number of observations of master's degrees and doctoral and professional degrees are low, but consistent with the overall finding that higher online rates lead to lower worker retention.

Calculating Degree Costs to the State

To estimate the cost of each degree, we use North Dakota University System's 2022-2023 Programs Completed report to determine the most popular degree by CIP code and degree level. We then create a plan of study for each degree and run the course selections through the most recent NDUS funding formula (NDUS, 2023).

Cost variations exist among the different programs at NDUS, particularly because of the numerous sub-CIP programs. We choose the most popular programs within each CIP to represent the general cost. The Base Rate and Credit Completion Factors (CCF), which tend to benefit smaller institutions, were considered, using the lowest rates from NDSU and UND for all degrees above associate degrees. For associate degrees, we use NDSCS rates.

Table A7

Cost of Degrees Offered by NDUS (In 2023 USD)

	Associate	Bachelor's	Master's	Doctoral
Agriculture	$17,191	$17,966	$16,678	$33,356
Natural Resources and Conservation	$13,517	$19,180	$16,678	$33,356
Architecture		$23,313	$20,541	
Area, Ethnic, Cultural, Gender, and Group Studies		$11,806	$9,071	
Communication and Journalism		$10,973	$8,778	$17,556
Communications Technologies		$10,973[†]		
Computer and Information Sciences	$21,237	$24,191	$23,408	$43,890
Culinary Service	$11,623			
Education	$13,201	$20,906	$16,678	$33,356
Engineering	$17,722	$25,661	$21,945	$43,890
Engineering Technologies	$12,484	$28,163	$21,945	
Foreign Language		$12,362		
Family and Consumer Sciences	$14,450	$16,744	$17,234	$33,356
Legal Professions and Studies		$12,684		$92,169
English Language and Literature		$11,411	$8,778	
General Studies and Humanities	$10,992	$10,973		
Biological and Biomedical Sciences		$16,620	$16,678	$33,356
Mathematics and Statistics		$13,460	$8,778	$13,167
Interdisciplinary Studies	$9,040	$15,230		
Parks, Recreation, Leisure, Fitness, and Kinesiology	$8,897	$19,794	$13,606	
Philosophy and Religious Studies		$18,851		
Physical Sciences		$18,163	$16,678	$33,356
Science Technologies		$19,977		
Psychology		$11,192	$8,778	$17,556
Homeland Security and Law Enforcement	$9,126	$11,850	$8,778	$17,556
Public Administration	$8,610	$15,969	$17,556	
Social Sciences		$12,121	$8,778	
Construction Trades	$41,470			
Mechanic and Repair Technologies	$40,896			
Precision Production	$38,600			
Transportation and Materials Moving		$18,975	$16,678	$33,356
Visual and Performing Arts	$10,059	$19,853	$8,778	$17,556
Health Professions and Related Programs	$18,080	$16,620	$43,012	$444,167
Business and Management	$16,000	$20,467	$16,678	
History		$11,192	$8,778	$17,556

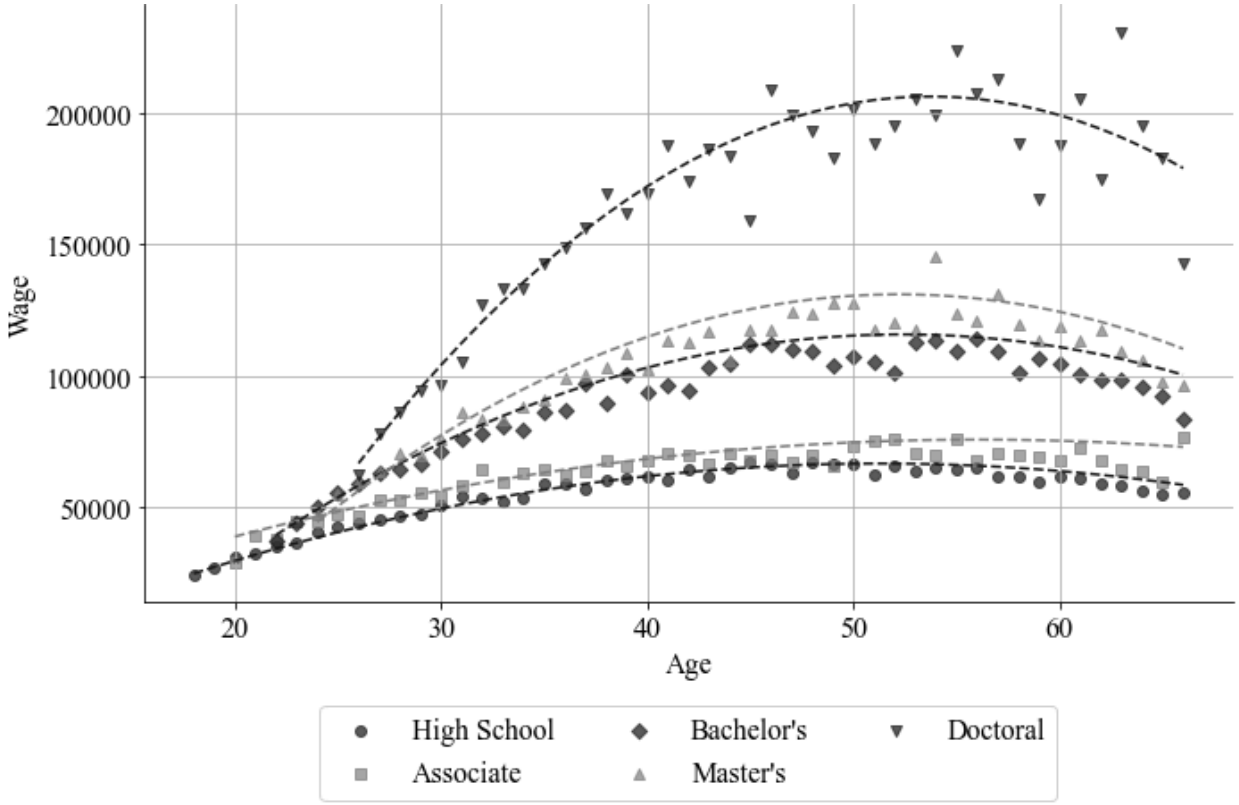
† This program ran for two years at UND under our period of study and a degree plan is no longer available. We used communication as a proxy for cost.

Marginal Wage Calculation

Figure A9 displays the average full-time wages in Minnesota at various levels of educational attainment and age for workers who work more than half of the year and who earn more than minimum wage (Census Bureau, 2024). We use Minnesota as a proxy to estimate the added wages at various degree levels for North Dakota's graduates over the span of their working years. North Dakota does not currently provide degree-specific post-secondary education wage estimates, but Minnesota does provide these estimates using a median (United States Census Bureau, n.d.). Additionally, we chose Minnesota due to the higher number of observations and lower year-by-year variance compared to North Dakota. For instance, the simple quadratic weighted least squares (WLS) regression maintained no statistical significance with only 395 observations compared to 3848 in Minnesota. For the early years of work, median wages are extremely close to mean wages, and help us provide conservative estimates of the benefits of education. The change in average wages in Minnesota were multiplied by the starting wages we attained from PSEO data (United States Census Bureau, 2024).

Figure A9

2018-2022 Annual Minnesota Full-Time Mean Wage by Degree Level and Quadratic Weighted Least Squares Regression (2023 USD)



ROI by CIP Codes

Table A8 displays the degree-specific MIRRs.

Table A8

Return on Investment for NDUS Funding (By CIP Codes, In 2023 USD)

CIP	Associate	Bachelor's	Master's	Doctoral
Agriculture	2.18%	11.82%	10.60%	
Natural Resources and Conservation		9.93%	12.86%	
Architecture		9.67%	11.14%	
Area, Ethnic, Cultural, Gender, and Group Studies		11.56%		
Communication and Journalism		12.12%	16.24%	
Communications Technologies		10.65%		
Computer and Information Sciences	10.70%	13.61%	12.72%	

Culinary Service	†			
Education		11.35%	13.83%	13.42%
Engineering		12.51%	9.94%	
Engineering Technologies	11.88%	12.28%		
Foreign Language		10.97%		
Family and Consumer Sciences	†	10.72%	12.48%	
Legal Professions and Studies				9.35%††
English Language and Literature		11.50%	13.75%	
General Studies and Humanities	†	11.54%		
Biological and Biomedical Sciences		9.88%		2.66%
Mathematics and Statistics		12.81%		
Interdisciplinary Studies		11.89%		
Parks, Recreation, Leisure, Fitness, and Kinesiology		9.25%	12.63%	
Philosophy and Religious Studies		9.93%		
Physical Sciences		10.71%		12.47%
Psychology		11.34%		11.38%
Homeland Security and Law Enforcement	12.22%	12.90%		
Public Administration	5.60%	11.57%	11.86%	
Social Sciences		11.97%		
Construction Trades	9.33%			
Mechanic and Repair Technologies	9.82%			
Precision Production	10.24%			
Transportation and Materials Moving		8.24%		
Visual and Performing Arts	†	9.09%	12.51%	
Health Professions and Related Programs	11.86%	14.53%	5.42%	5.92%††
Business and Management	9.41%	12.81%	14.44%	

† These degrees either failed to earn wages higher than high school wages for every year of data or returned a negative MIRR. They were used to calculate the pooled MIRR for the degree level, but their negative values are not suggestive for CIP-level data.

†† For legal and medical, due to the rarity of a master's degree preceding these professional degrees, the increase in wages is in relation to bachelor's degrees. Additionally, we are not assuming any reduced credits due to graduate credit transfers. In North Dakota and Minnesota, the most common bachelor's degree among working healthcare doctoral degree holders is in biology rather than health professions (a registered nurse).